



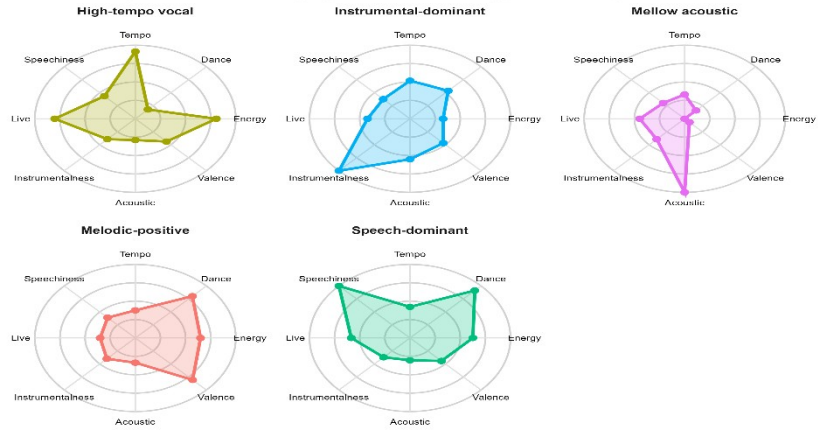
Acoustic Structures, Not Features:

Explaining HSS Inconsistency with PCA-Based Clusters

(Figure 1)

Acoustic feature profiles by cluster

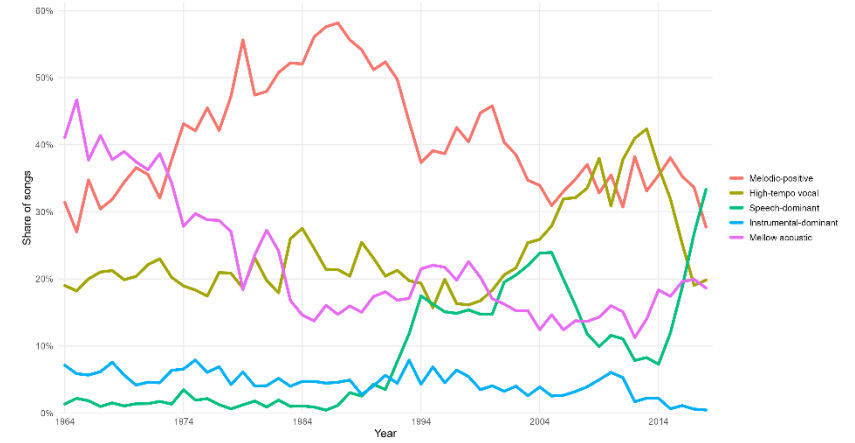
Points show average feature levels for each group. Extreme values are capped for readability.



(Figure 2)

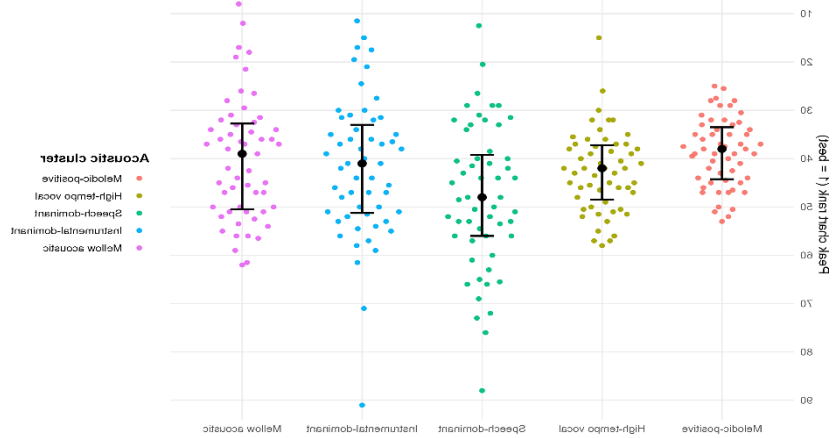
Cluster prevalence over time

Yearly share of charting songs



Peak chart rank by acoustic structure

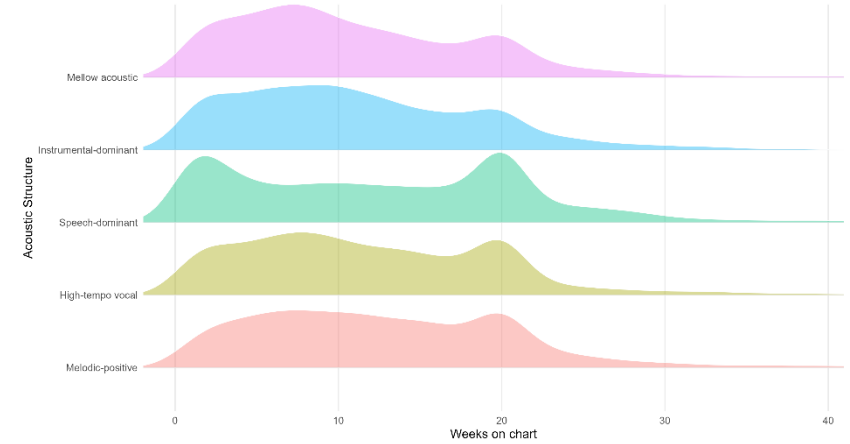
Points are yearly medians (entry year). Black dot shows the median across years; black bar shows the IQR (25th–75th percentile).



(Figure 3)

Acoustic Structure Chart Longevity

Ridgeline densities of song chart duration (trimmed at 99th percentile)



(Figure 4)

1. Knowledge Building

1.1 Topic and Question

This visualisation addresses a persistent and well-documented limitation in the Hit Song Science (HSS) literature: the inconsistent and often contradictory relationships reported between song-level acoustic features and musical success. Across studies, individual features such as tempo, energy, loudness, or danceability are frequently used as independent predictors, yet their estimated effects vary substantially across datasets, chart systems, time periods, and outcome definitions, and are often weak or unstable (Pachet & Roy, 2008; Interiano et al., 2018; Seufitelli et al., 2023; Yap & Raheem, 2024). This instability has led to scepticism regarding the explanatory value of acoustic features and raised concerns that they may proxy genre, production practices, or stylistic conventions rather than intrinsic musical properties (Downie, 2003; Pachet & Roy, 2008).

The core question motivating this visualisation is whether this variability can be explained by the fact that acoustic features do not operate independently, but instead combine into higher-level, genre-like acoustic structures whose prevalence and relationship with success change over time. Rather than asking whether specific features predict success, the visualisation investigates whether dimensionality reduction reveals coherent acoustic regimes, whether these regimes exhibit temporal dynamics, and whether their associations differ depending on whether success is defined by peak chart performance or by chart persistence. This reframing responds to calls within HSS for structure-aware, explanatory approaches that move beyond feature-level prediction (Seufitelli et al., 2023) and aligns with accounts of musical success as cumulative and socially reinforced (Salganik et al., 2006; Rosati et al., 2021).

1.2 Composite Visualisation

Figure 1 shows acoustic feature profiles of clusters identified via principal component analysis. Each cluster forms a coherent and interpretable configuration of features corresponding to recognisable stylistic regimes, such as speech-dominant tracks with elevated speechiness and danceability, mellow acoustic tracks with high acousticness and low energy, and instrumental-dominant tracks characterised by high instrumentality. The semantic clarity of these profiles supports claims that engineered acoustic features primarily encode genre and production conventions rather than isolated musical attributes (Downie, 2003; Pachet & Roy, 2008), demonstrating that PCA recovers meaningful structure from high-dimensional acoustic data.

Figure 2 presents the temporal prevalence of these acoustic structures as yearly shares of charting songs. Cluster dominance varies substantially over time, indicating non-stationary stylistic regimes rather than stable acoustic distributions. This historical contingency implies that studies pooling long time horizons average across heterogeneous musical eras, providing a plausible explanation for inconsistent acoustic effects reported across datasets and periods (Interiano et al., 2018; Seufitelli et al., 2023).

Figure 3 examines the relationship between acoustic structure and peak chart rank. Median peak positions differ modestly across clusters, but substantial overlap remains, indicating that no acoustic structure consistently dominates early chart success. This aligns with findings that acoustic features have limited explanatory power for peak-based success, reinforcing critiques of peak rank as a noisy proxy for initial exposure rather than durable popularity (Pachet & Roy, 2008; Lee & Lee, 2018).

Figure 4 shows distributions of weeks on chart by acoustic structure, where clearer differences emerge in distributional shape and tail behaviour, with some clusters exhibiting greater persistence. This suggests that acoustic structure conditions durability rather than initial selection, consistent with process-based accounts emphasising cumulative advantage and post-entry social reinforcement (Salganik et al., 2006; Rosati et al., 2021).

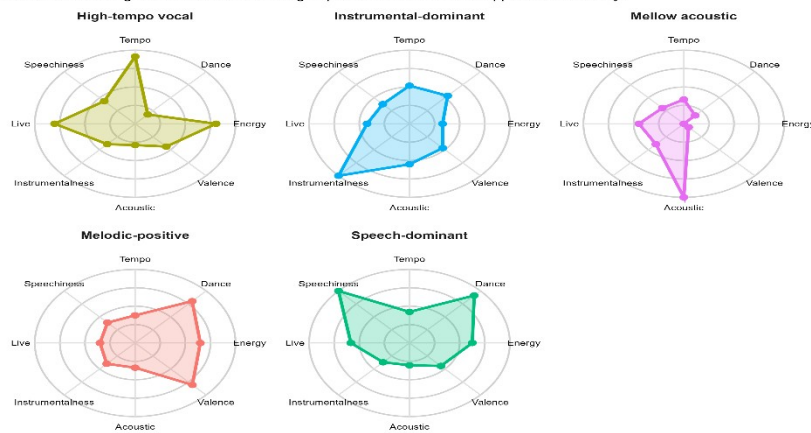
Taken together, the composite visualisation provides explanatory insight unavailable from any single panel. While individual acoustic features appear weak and unstable when analysed in isolation, grouping them into acoustic structures reveals that their relevance depends on temporal context and on how success is operationalised. The visualisation shows that acoustic characteristics primarily differentiate persistence rather than peak performance, and that shifting stylistic regimes help explain inconsistent findings in prior HSS research, supporting a move from feature-level prediction towards structure-aware, context-sensitive explanation (Seufitelli et al., 2023; Yap & Raheem, 2024).

2. Theoretical Frameworks

This section focuses on Figure 1, which examines how acoustic characteristics structure popular music and whether coherent acoustic patterns emerge in large-scale music data. The design was guided by the ASSERT framework as a structured approach to purposeful visualisation, alongside a Grammar of Graphics (GoG) perspective that decomposes the figure into data, aesthetic mappings, geometries, and scales rather than relying on predefined chart types (Ferster & Shneiderman, 2012; Wilkinson, 2005).

Acoustic feature profiles by cluster

Points show average feature levels for each group. Extreme values are capped for readability.



Ask

The process began by defining the analytical question, audience, and task, consistent with guidance that effective visual analysis should be question-driven rather than chart-driven (Few, 2009; Few, 2012; Kirk, 2019). The question responds to inconsistent findings in the HSS literature on the role of acoustic features, often attributed to heterogeneous samples, differing chart systems, and an emphasis on prediction over explanation (Seufitelli et al., 2023). Restricting analysis to song-level acoustic characteristics within a single historical window and chart context reduces these sources of heterogeneity and establishes a coherent empirical basis for analysis.

Search for Information

Following Few's (2009) criteria for meaningful data, a publicly available dataset with transparent provenance, prior use in HSS research, and clear documentation was selected (Silva et al., 2019). The data comprise over 20,000 song-level observations spanning 1958–2018 and are consistently defined, multivariate, and historically complete, supporting

analysis of long-run patterns and complex acoustic relationships. Dataset selection was treated as a prerequisite to visual encoding, consistent with task- and data-first design models that position data abstraction before visual form (Few, 2009; Few, 2012; Munzner, 2009).

Structuring the Data

Data were structured prior to visualisation to support multivariate analysis and Grammar of Graphics–based encoding (Munzner, 2009). Song-level acoustic features were treated as quantitative variables and cluster membership as categorical, reflecting that variable type constrains valid aesthetic mappings (Wilkinson, 2005). Acoustic features were standardised, and dimensionality reduced using PCA to ensure comparability across scales, capture dominant variance structure, and mitigate collinearity prior to clustering (Cleveland & McGill, 1984; James et al., 2013; Johnson & Wichern, 1998). For visualisation, cluster-level mean standardised feature values were used to summarise central tendencies and foreground structural differences, with fixed radial limits applied consistently across facets to preserve perceptual comparability (Ware, 2012).

Envision the Answer

Given the exploratory nature of the analysis, the focus was on specifying what would constitute a meaningful explanation rather than a predetermined outcome. This follows Bates' Berry-picking Model, which frames information seeking as an iterative, non-linear process in which analytic goals evolve as new evidence is encountered (Lillard & Ha, 2015). Multiple analytical perspectives—PCA projections, loadings, and cluster-level summaries—were examined in parallel to corroborate emerging patterns across complementary views, consistent with models of visual analysis that emphasise iterative sense-making over optimisation of a single display (Ferster & Shneiderman, 2012; Munzner, 2009). Comparing patterns across representations helped distinguish structure driven by extreme values from that reflecting central tendencies.

Represent the Data

To represent the high-dimensional acoustic feature space, the visualisation was designed to support comparison of multivariate cluster profiles while also situating these profiles within the broader variance structure of the data. Radar plots were used to compare relative feature configurations across clusters, supporting exploratory pattern detection in multivariate settings (Ware, 2012), while PCA projections were examined separately to assess global structure in reduced-dimensional space (David & Tukey, 1977; Munzner, 2009).

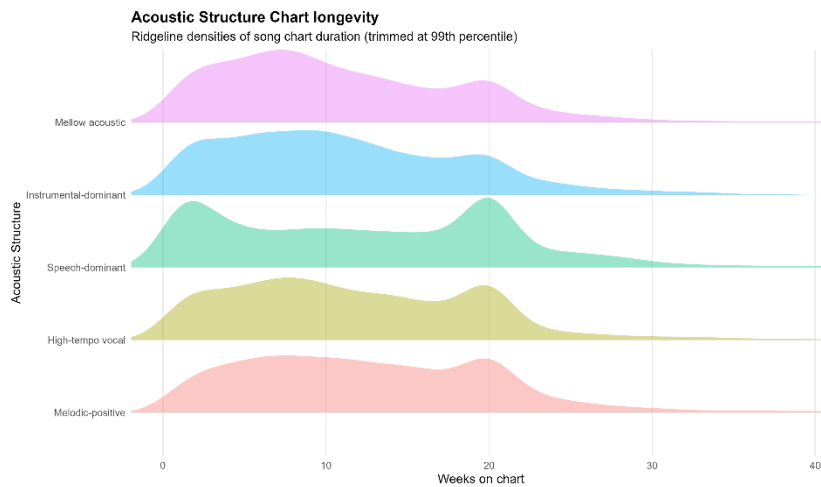
Following the GoG, feature identity was encoded by angular position and standardised feature magnitude by radial distance in polar coordinates, with orientation held constant across facets to preserve perceptual comparability (Wilkinson, 2005; Cleveland & McGill, 1984). Holistic cluster profiles were encoded using layered polygons and paths, separating contextual from focal structure through muted base layers and coloured overlays (Kirk, 2019). Defining features were highlighted using a fixed threshold ($|\text{mean } z| \geq 0.6$), while fixed radial limits with explicit clipping preserved overall structure and improved perceptual resolution and transparency (Few, 2009; Ware, 2012).

Tell a Story

Building on these encodings, the visualisation followed an overview-to-detail narrative, presenting global structure before progressively more focused views to guide comparison and interpretation (Shneiderman, 1996). Global acoustic structure was introduced using PCA projections, followed by cluster-level acoustic profiles and feature-level summaries. Quantitative magnitude was encoded using position to support accurate comparison, while colour differentiated cluster membership. Muted reference layers provided contextual baselines, with shape and transparency used sparingly to de-emphasise background elements and signal clipped values (Few, 2009; Ware, 2012).

This narrative structure supports interpretation of clusters as acoustic archetypes rather than discrete genres, guiding comparison while avoiding over-categorisation or false precision (Segel & Heer, 2010). By foregrounding dominant features and relative differences, the visualisation enables readers to identify salient patterns and compare cluster profiles directionally without reliance on raw numerical tables.

3. Accessibility



Accessibility in visualisation concerns how effectively a figure supports perception and interpretation across diverse abilities and contexts, shaped by perceptual and cognitive constraints (Ware, 2012; Munzner, 2014). For the creation of Figure X, accessibility was treated as task- and audience-relative (Kirk, 2012). Designed for data-literate but non-specialist readers, the figure prioritises comparison of distributional patterns in chart longevity over precise value extraction, aligning with guidance that visual encodings should match analytical tasks rather than maximise numerical precision (Munzner, 2014; Few, 2009).

In selecting an appropriate visual representation, several mapping options were considered. Boxplots were rejected because their quartile-based summaries assume statistical familiarity, increasing cognitive effort and obscuring distributional structure for non-specialist audiences (Few, 2009). ECDFs were also rejected, as slope-based and cumulative encodings make cross-series differences less immediately perceptible and increase inference burden in comparative tasks (Heer et al., 2010; Munzner, 2014). Ridgeline density plots were therefore selected because they foreground distributional shape, making differences in spread, skewness, and multimodality visually salient (Wilkinson, 2005).

Layout and encoding decisions were similarly guided by perceptual accessibility. Chart longevity was mapped to a shared axis, with distributions separated vertically rather than overlapping. This reduces visual crowding and attentional competition, supporting discrimination between groups during parallel processing (Healey & Enns, 2011), while reinforcing position-based comparison by exploiting spatial position as a high-precision perceptual channel and avoiding reliance on weaker encodings such as colour for magnitude judgement (Munzner, 2014). Each ridgeline encodes a single distribution using one geometry,

avoiding competing visual signals that could interfere with pattern recognition and attentional processing (Ware, 2012).

Scaling decisions were informed by similar perceptual considerations. Data scaling was not applied to avoid perceptual distortion from rescaled axes, and log transformations were rejected because they substantially altered distributional shape, undermining interpretability. Consistent with Tufte's (1985) emphasis on preserving perceptual resolution and Ware's (2012) discussion of perceptual limits, extreme values were trimmed at the 99th percentile to prevent long tails from compressing central structure. This improves visibility of typical patterns while necessarily de-emphasising rare extremes.

Labelling decisions were guided by cognitive load considerations. Direct labelling on the y-axis removes the need for legend lookups, reducing working-memory demands during comparison and supporting more efficient visual scanning (Ware, 2012; Few, 2009; Munzner, 2014). Together, these design choices prioritise perceptual clarity and structural inference over precise value extraction, aligning the visual encoding with the comparative and exploratory task of the figure.

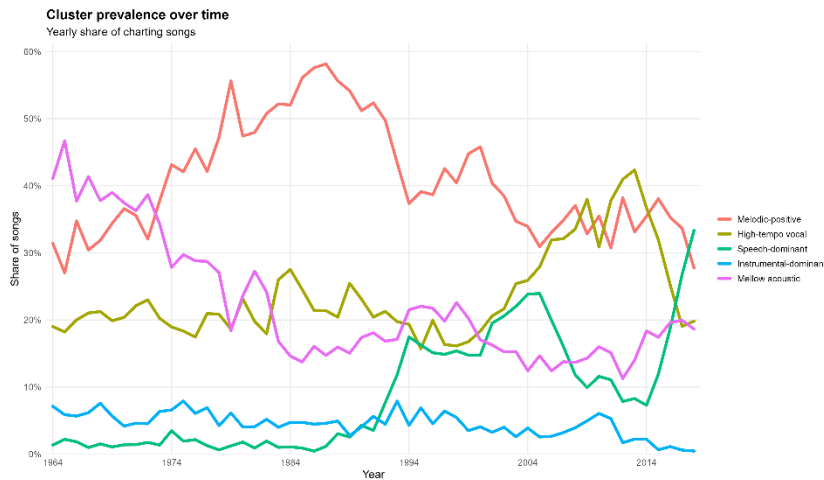
Several accessibility limitations nevertheless warrant consideration. Category separation relies primarily on colour, which limits accessibility for colour-vision-deficient users and under greyscale reproduction. Although direct labelling provides a non-colour cue, reliance on text introduces additional accessibility demands related to legibility and reading ability (Ware, 2012; Few, 2009). Accessibility could be improved by introducing redundant non-textual cues, such as outline variation or increased luminance contrast, in line with guidance to avoid reliance on a single perceptual channel (Ware, 2012; Munzner, 2014).

Interpretation of the ridgeline densities is also schema-dependent, as correct reading requires familiarity with kernel density representations and probabilistic encodings. While statistically literate viewers can efficiently extract distributional structure, less experienced audiences may misinterpret ridge height or area as raw counts, increasing cognitive effort (Yau, 2013; Munzner, 2014; Kirk, 2012). This dependence could be mitigated through light interpretive scaffolding, such as a brief caption explaining density interpretation or a single annotated reference, without altering the core visual encoding.

Finally, as a static visual output, the figure lacks a non-visual alternative such as a textual summary or tabular representation of key distributional properties. This limits

accessibility for screen-reader users and places greater interpretive burden on viewers unfamiliar with density-based encodings, contrary to guidance recommending complementary text-based explanations for visual content (Few, 2009; Munzner, 2014). Pairing the figure with a small summary table or reference markers would allow it to retain its pattern-focused role while also supporting exact value extraction.

4. Visualisation Choice



4.1 Justification of the Type of Visualisation

The aim of this visualisation is to show the presence of each acoustic cluster on the charts and how that presence varies over time, providing temporal context for variation in acoustic feature effects reported in HSS research across datasets (Seufitelli et al., 2023). A multi-series line chart was selected for its suitability in representing temporal change, making it appropriate for analysing variation in cluster prevalence across decades (Few, 2009; Wilkinson, 2005).

Plotting all clusters on a shared temporal axis enables direct comparison of trends, slopes, persistence, and turning points, which is central to identifying long-run shifts in stylistic dominance (Cleveland & McGill, 1984). Visual crossings and divergence between lines make both structural change and volatility explicit, supporting interpretation of long-term dynamics alongside short-term year-to-year fluctuations (Ware, 2012). Cluster prevalence is encoded as yearly shares rather than raw counts to control for variation in the total number of charting songs, enabling meaningful comparison across time periods (Cairo, 2012; Tufte, 1985). By situating acoustic clusters within their changing prevalence over time, the figure provides a contextual explanation for why relationships between acoustic features and chart success may differ across studies (Salganik et al., 2006). Feature-level results can therefore be interpreted relative to broader stylistic composition rather than as time-invariant effects.

The visualisation has recognised limitations. Multiple overlapping lines can reduce discriminability, particularly when cluster shares converge, and the chart prioritises trend comparison over precise snapshot composition at individual time points (Healey & Enns, 2011; Ware, 2012).

Interpretation also requires legend lookup, introducing minor cognitive overhead, and proportions may be misread if readers overlook that values represent shares rather than counts (Few, 2009; Cairo, 2012). These limitations were accepted because the primary analytical task is comparison of temporal trends rather than detailed composition at a single point in time (Munzner, 2009).

4.2 Alternatives

Sankey Diagram

A Sankey diagram was considered because it is commonly used to visualise flows between categories and could, in principle, map acoustic clusters across time periods (Heer et al., 2010). Such diagrams are visually engaging and often accessible to non-technical audiences (Kirk, 2019). However, the flow metaphor implies transitions or redistribution between categories, whereas cluster membership is fixed and only prevalence changes over time (Munzner, 2014). Using a Sankey diagram would therefore misrepresent the data structure (Tufte, 1985).

In addition, Sankey diagrams require aggregation into coarse temporal bins (e.g. decades) to remain legible, obscuring year-to-year variation and short-lived dynamics (Few, 2009). Quantitative comparison via flow thickness is perceptually weaker than comparison along a shared positional axis, making relative cluster prevalence harder to assess (Cleveland & McGill, 1984). For these reasons, the Sankey diagram was rejected in favour of a multi-series time-series line chart, which more accurately represents changing prevalence over continuous time and directly supports the analytical task (Munzner, 2009).

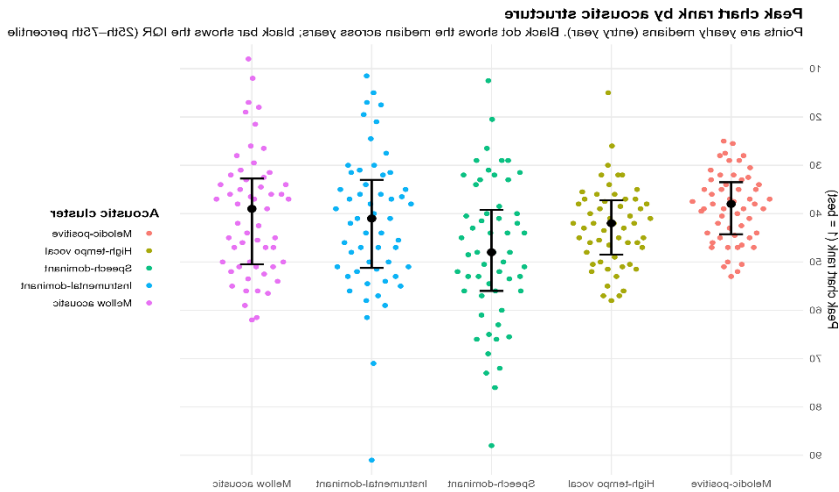
Waffle Chart

A waffle chart was considered effective for communicating proportional composition at a single point in time (Few, 2012). Its discrete grid metaphor is intuitive and accessible, making it well suited to presenting composition for a single year or an aggregated period (Cairo, 2012). However, waffle charts do not scale to long time series without extensive repetition or animation (Few, 2009). Applied to data spanning **1960–2018**, this would fragment the narrative and obscure temporal continuity (Munzner, 2014). Discrete grids also make it difficult to perceive rates of change, crossovers, and relative trajectories, weakening comparison over time (Cleveland & McGill, 1984). Faceting across many periods would

further reduce comparability, while aggregation to broader intervals (e.g. decades) would conceal short-term dynamics (David & Tukey, 1977).

Because the analytical goal is to examine variation in cluster presence over time rather than static composition, the waffle chart was rejected as unsuitable for this analysis (Munzner, 2009).

5. Implications and Improvements



5.1 Ethical Implications

Figure 3 compares peak chart rank across acoustic clusters using yearly medians, which may invite audiences to infer that certain clusters are inherently more successful than others. Although the visualisation is descriptive, research in information visualisation shows that grouped summary statistics are frequently interpreted causally, even when no causal claim is intended (Correll & Gleicher, 2014; Tufte, 1985). The juxtaposition of cluster-level medians and interquartile ranges may therefore encourage viewers to attribute performance differences directly to acoustic structure.

By foregrounding central tendencies, the beeswarm-style encoding makes between-cluster differences salient while substantial within-cluster variability is visually secondary. Work on statistical cognition suggests that audiences often overweight aggregate summaries relative to overlap and dispersion when comparing categories (Gigerenzer & Edwards, 2003). In this figure, considerable overlap in yearly medians and interquartile ranges indicates that differences in peak performance are weak and inconsistent; nevertheless, the prominence of median markers may still encourage overinterpretation of modest effects.

More critically, because clusters are defined solely by acoustic similarity, the figure risks reinforcing content-deterministic narratives of musical success that marginalise the roles of structural inequality, industry gatekeeping, and cumulative advantage in cultural markets (Salganik et al., 2006; Ferreira & Waldfogel, 2012). This concern is well established in HSS research. Large-scale studies show that popularity cannot be reliably predicted from acoustic features alone, and that early claims of “hit prediction” were often overstated or failed to

generalise (Pachet & Roy, 2008; Salganik et al., 2006). Reviews further emphasise that exposure, social influence, and cumulative advantage dominate chart outcomes, with acoustic characteristics playing at most a secondary role (Ferreira & Waldfogel, 2012; Seufitelli et al., 2023).

In addition, the use of yearly medians masks extreme outcomes, potentially downplaying the influence of rare but consequential events such as breakout hits or heavily promoted releases. Aggregation choices are known to shape narrative interpretation in visual analytics and can obscure mechanisms operating in the tails of distributions (Cairo, 2012). Without explicit annotation, viewers may also overlook differences in the number of observations or years contributing to each cluster, misinterpreting similar point densities as equivalent evidential support (Munzner, 2014).

Taken together, these factors mean that, while the figure is effective for comparative exploration, it requires careful framing. Without this, it may inadvertently reinforce reductionist interpretations that treat acoustic structure as a sufficient explanation for chart success, contrary to the consensus of contemporary HSS research (Salganik et al., 2006; Pachet & Roy, 2008; Seufitelli et al., 2023).

5.2 Future Improvements

Several extensions could enhance both clarity and analytical scope. Interactive versions of the figure could allow users to toggle between yearly medians, alternative summaries (e.g. trimmed means), and raw song-level peak ranks, making aggregation choices explicit. Prior work shows that interactivity supports critical interrogation of summaries by enabling exploration of uncertainty and alternative representations (Heer & Shneiderman, 2012). Because such decisions are analyst-driven, ethical responsibility lies in ensuring that design choices do not obscure mechanisms or overstate evidential support (Cairo, 2012; Correll & Gleicher, 2014). However, the present work is limited to static visualisations due to technical and time constraints.

Incorporating industry-level variables—such as label affiliation, promotional intensity, or playlist exposure—would better align the visualisation with established explanations of popularity that emphasise institutional and social mechanisms over purely acoustic content (Salganik et al., 2006; Ferreira & Waldfogel, 2012). The absence of such variables constrained the current analysis to acoustic features alone.

Conditioning peak rank on historical context (e.g. decade or chart regime) would further account for structural changes in chart dynamics and listening behaviour, improving comparability across eras (Ferreira & Waldfogel, 2012). This extension was not pursued to maintain focus on the primary analytical objective.

Finally, accessibility and transparency could be improved through explicit sample-size annotation, colour-blind-safe palettes, and complementary summaries that foreground uncertainty and overlap. These additions would reduce the risk of overstating evidential support and clarify the limits of the data—core ethical responsibilities in visualisation design rather than aesthetic refinements (Cairo, 2012).

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Appendices

Appendix A

Project Code Files

```
# R/01_data_cleaning.R -----
# Song-level dataset for clustering + popularity/persistence analyses.
# Output: clean/song_df.csv

suppressPackageStartupMessages({
  library(dplyr)
  library(readr)
})

source("R/helpers.R")
ensure_dir("clean")

# ---- Load inputs -----

acoustic_features <- read_tsv("raw/features/acoustic_features.csv", show_col_types =
FALSE)
song_chart      <- read_tsv("raw/popularity/song_chart.csv", show_col_types = FALSE)
songs           <- read_tsv("raw/metadata/songs.csv", show_col_types = FALSE)

# ---- Prepare chart table (recode + derive year) -----

song_chart_recoded <- song_chart |>
  transmute(
    song_id,
    week = as.Date(week), # format like "1989-09-09"
    year = as.integer(format(as.Date(week), "%Y")),
    weeks_on_chart = if_else(weeks_on_chart == 0, 1L, as.integer(weeks_on_chart)),
    rank_score
  )

# ---- Restrict to songs observed from chart entry -----
# Keep songs whose first observed weeks_on_chart is 1 (avoids left-truncated runs)

song_chart_recoded <- song_chart_recoded |>
  group_by(song_id) |>
  filter(min(weeks_on_chart, na.rm = TRUE) == 1L) |>
  ungroup()

# ---- Compact song-year presence info (for Figure 2) -----

song_year_info <- song_chart_recoded |>
  distinct(song_id, year) |>
  group_by(song_id) |>
```

```

summarise(
  entry_year = min(year, na.rm = TRUE),
  exit_year = max(year, na.rm = TRUE),
  n_years_on_chart = n(),
  years_on_chart = paste(sort(unique(year)), collapse = ";"),
  .groups = "drop"
)

```

---- Song-level chart summary -----

```

song_df <- song_chart_recoded |>
  group_by(song_id) |>
  arrange(weeks_on_chart, week) |>
  mutate(
    entry_week = first(week),
    exit_week = last(week),
    song_rank_entry = first(rank_score),
    song_rank_peak = max(rank_score, na.rm = TRUE)
  ) |>
  slice_tail(n = 1) |>
  ungroup() |>
  rename(song_rank_final = rank_score) |>
  left_join(
    songs |>
      select(song_id, song_name, song_type, explicit),
    by = "song_id"
  ) |>
  left_join(
    acoustic_features |>
      select(
        song_id, duration_ms,
        acousticness, danceability, energy, instrumentalness,
        liveness, loudness, speechiness, valence, tempo
      ) |>
      filter(tempo != 0),
    by = "song_id"
  ) |>
  left_join(
    song_year_info,
    by = "song_id"
  ) |>
  select(
    song_id, song_name, song_type, explicit,
    entry_year, exit_year, n_years_on_chart, years_on_chart,
    weeks_on_chart,
    song_rank_entry, song_rank_peak, song_rank_final,
    entry_week, exit_week,
    duration_ms,
    acousticness, danceability, energy, instrumentalness,

```

```

    liveness, loudness, speechiness, valence, tempo
  ) |>
  filter(if_all(everything(), ~ !is.na(.)))

write_csv(song_df, "clean/song_df.csv")
message("Data cleaning complete: clean/song_df.csv")

```

```

# R/02_eda.R -----
# EDA:
# - Data integrity checks
# - PCA fitting and diagnostics
# - Cluster selection and sensitivity checks
# - Final k-means solution
# - Cluster summaries for justification and figures

suppressPackageStartupMessages({
  library(dplyr)
  library(readr)
  library(tidyr)
  library(ggplot2)
  library(cluster)
})

source("R/helpers.R")

DIR_EDA <- "outputs/eda"
DIR_EDA_TABLES <- file.path(DIR_EDA, "tables")

ensure_dir(DIR_EDA)
ensure_dir(DIR_EDA_TABLES)

song_df <- read_csv("clean/song_df.csv", show_col_types = FALSE)

# ---- Local helper: PCA + k-means projection (EDA only) -----

make_pca_plot <- function(df, vars, k_clusters, title, out_file) {
  X <- df %>% select(all_of(vars)) %>% as.data.frame()
  pca <- prcomp(X, center = TRUE, scale. = TRUE)

  k_pcs <- min(K_PCS_CLUSTER, ncol(pca$x))
  pc_scores <- as.data.frame(pca$x[, 1:k_pcs, drop = FALSE])

  km <- run_kmeans(pc_scores, k = k_clusters)

  plot_df <- as.data.frame(pca$x[, 1:2, drop = FALSE])
  names(plot_df) <- c("PC1", "PC2")
}

```

```

ve <- (pca$sdev^2) / sum(pca$sdev^2)

plot_df$cluster <- factor(km$cluster)
plot_df <- apply_cluster_labels(plot_df)

p <- ggplot(plot_df, aes(PC1, PC2, colour = cluster)) +
  geom_point(alpha = 0.55, size = 1.1) +
  labs(
    title = title,
    x = sprintf("PC1 (%.1f%%)", 100 * ve[1]),
    y = sprintf("PC2 (%.1f%%)", 100 * ve[2])
  ) +
  scale_colour_cluster() +
  theme_project()

save_fig(p, out_file, width = 9.5, height = 6.5, dpi = 300)
invisible(p)
}

# ---- Run EDA -----

with_log(file.path(DIR_EDA, "eda_log.txt"), {

  log_h1("EDA")

  # -----
  # 01) Data integrity
  # -----

  log_h2("01) Data integrity")

  eda_01_integrity <- song_df %>%
    summarise(
      n_songs = n(),
      missing_any = sum(!complete.cases(.)),
      min_entry_year = min(entry_year, na.rm = TRUE),
      max_exit_year = max(exit_year, na.rm = TRUE),
      min_weeks_on_chart = min(weeks_on_chart, na.rm = TRUE),
      max_weeks_on_chart = max(weeks_on_chart, na.rm = TRUE),
      min_peak_rank = min(song_rank_peak, na.rm = TRUE),
      max_peak_rank = max(song_rank_peak, na.rm = TRUE)
    )

  log_table(eda_01_integrity, digits = 0)

  stopifnot(anyDuplicated(song_df$song_id) == 0)
  stopifnot(all(song_df$weeks_on_chart >= 1))
  stopifnot(all(song_df$entry_week <= song_df$exit_week))
  stopifnot(all(song_df$entry_year <= song_df$exit_year))

```



```

log_h2("01B) Popularity summary")

eda_01b_pop <- song_df %>%
  summarise(
    across(
      all_of(pop_vars),
      list(
        min = ~ min(.x, na.rm = TRUE),
        med = ~ median(.x, na.rm = TRUE),
        p99 = ~ as.numeric(quantile(.x, 0.99, na.rm = TRUE)),
        max = ~ max(.x, na.rm = TRUE)
      ),
      .names = "{.col}__{.fn}"
    )
  ) %>%
  pivot_longer(everything(), names_to = "stat", values_to = "value")

log_table(eda_01b_pop, digits = 3)

# -----
# 02) Core PCA
# -----

log_h2("02) PCA (core features)")

pca_out <- fit_pca_core(song_df, vars = vars_core, k_pcs = K_PCS_CLUSTER)

pca_core <- pca_out$pca
ve <- pca_out$ve
cum_ve <- pca_out$cum_ve
pc_scores <- pca_out$scores

eda_02_var <- tibble(
  PC = paste0("PC", seq_along(ve)),
  var_explained = ve,
  cum_var = cum_ve
)

log_table(eda_02_var, digits = 3)

saveRDS(pca_core, "clean/pca_core.rds")
write_eda_table(eda_02_var, "clean/pca_variance.csv", digits = 6)

loadings_df <- as.data.frame(pca_core$rotation) %>%
  tibble::rownames_to_column("feature") %>%
  pivot_longer(-feature, names_to = "PC", values_to = "loading")

write_eda_table(loadings_df, "clean/pca_loadings.csv", digits = 6)

# -----

```

```
# 03) Cluster diagnostics
```

```
# -----
```

```
log_h2("03) Cluster diagnostics")
```

```
elbow_tbl <- tibble(  
  k = K_RANGE_DIAG,  
  tot_withinss = sapply(K_RANGE_DIAG, function(k) {  
    run_kmeans(pc_scores, k = k)$tot.withinss  
  })  
)
```

```
log_table(elbow_tbl, digits = 2)
```

```
set.seed(SEED)  
n_sil <- min(N_SIL_SAMPLE, nrow(pc_scores))  
idx <- sample(seq_len(nrow(pc_scores)), n_sil)  
d_sub <- dist(pc_scores[idx, , drop = FALSE])
```

```
sil_tbl <- tibble(  
  k = K_RANGE_DIAG,  
  avg_sil_width = sapply(K_RANGE_DIAG, function(k) {  
    km <- run_kmeans(pc_scores[idx, , drop = FALSE], k = k)  
    mean(silhouette(km$cluster, d_sub)[, "sil_width"])  
  })  
)
```

```
log_table(sil_tbl, digits = 3)
```

```
write_eda_table(  
  left_join(elbow_tbl, sil_tbl, by = "k"),  
  file.path(DIR_EDA_TABLES, "03_cluster_diagnostics_elbow_silhouette.csv"),  
  digits = 6  
)
```

```
# -----
```

```
# 04) PCA sensitivity checks
```

```
# -----
```

```
log_h2("04) PCA sensitivity")
```

```
make_pca_plot(  
  song_df, vars_core, K_CLUSTERS_FINAL,  
  "PCA clusters (excluding loudness)",  
  file.path(DIR_EDA, "eda_pca_clusters_no_loudness.png")  
)
```

```
make_pca_plot(  
  song_df, c(vars_core, "loudness"), K_CLUSTERS_FINAL,  
  "PCA clusters (including loudness)",
```

```

file.path(DIR_EDA, "eda_pca_clusters_with_loudness.png")
)

make_pca_plot(
  song_df, setdiff(vars_core, "tempo"), K_CLUSTERS_FINAL,
  "PCA clusters (excluding tempo)",
  file.path(DIR_EDA, "eda_pca_no_tempo.png")
)

# -----
# 05) Final clustering
# -----

log_h2("05) Final cluster solution")

km_final <- run_kmeans(pc_scores, k = K_CLUSTERS_FINAL)

eda_df <- song_df %>%
  select(all_of(vars_core)) %>%
  mutate(cluster = factor(km_final$cluster)) %>%
  apply_cluster_labels()

log_table(count(eda_df, cluster), digits = 0)

# ---- Cluster summaries (EDA justification) -----

eda_df_scaled <- eda_df %>%
  mutate(across(all_of(vars_core), ~ as.numeric(scale(.x))))

write_eda_table(
  eda_df_scaled %>%
    group_by(cluster) %>%
    summarise(across(all_of(vars_core), mean), .groups = "drop"),
  file.path(DIR_EDA_TABLES, "05b_cluster_profiles_mean_z.csv"),
  digits = 3
)

write_eda_table(
  eda_df %>%
    group_by(cluster) %>%
    summarise(across(all_of(vars_core), median), .groups = "drop"),
  file.path(DIR_EDA_TABLES, "05c_cluster_profiles_medians.csv"),
  digits = 3
)

log_table(
  as.data.frame(pc_scores) %>%
    mutate(cluster = factor(km_final$cluster)) %>%
    apply_cluster_labels() %>%
    group_by(cluster) %>%

```

```

    summarise(across(where(is.numeric), mean), .groups = "drop"),
    digits = 3
  )

  assign("km_final", km_final, envir = .GlobalEnv)
})

# -----
# 06) Materialise clustered data
# -----

song_df_clustered <- song_df %>%
  mutate(cluster = factor(km_final$cluster)) %>%
  apply_cluster_labels()

write_csv(song_df_clustered, "clean/song_df_clustered.csv")

# -----
# 07) Cluster feature profiles (long; figures)
# -----

cluster_profile_df <- eda_df_scaled %>%
  pivot_longer(
    cols = all_of(vars_core),
    names_to = "feature",
    values_to = "value_z"
  ) %>%
  group_by(cluster, feature) %>%
  summarise(mean_z = mean(value_z, na.rm = TRUE), .groups = "drop")

write_csv(cluster_profile_df, "clean/cluster_feature_profiles.csv")

message("EDA complete.")

```

```

# R/03_build_figures.R -----
# Purpose:
# - Orchestrate figure generation (no figure code lives here).
# - Load shared inputs once (clustered data + exported PCA artefacts).
# - Define figure-wide constants used across figure scripts.
# - Source individual figure scripts in a fixed order.

suppressPackageStartupMessages({
  library(tidyverse)
  library(zoo)
  library(ggribes)

```

```

library(patchwork)
library(ggbeeswarm)
})

source("R/helpers.R")

DIR_FIG <- "outputs/figures"
ensure_dir(DIR_FIG)

# ---- Load shared inputs -----

song_df <- readr::read_csv("clean/song_df_clustered.csv", show_col_types = FALSE)

pca_core <- readRDS("clean/pca_core.rds")

pca_variance <- readr::read_csv("clean/pca_variance.csv", show_col_types = FALSE)
pca_loadings <- readr::read_csv("clean/pca_loadings.csv", show_col_types = FALSE)

cluster_profile_df <- read_csv("clean/cluster_feature_profiles.csv", show_col_types =
FALSE)

# ---- Figure-wide constants -----
# Keep figure scripts free of hard-coded settings.

K_PCS_CLUSTER <- K_PCS_CLUSTER # clustering space (EDA-locked; helpers.R)
K_PCS_PLOT   <- 2             # default PCA projection (PC1–PC2)

BASE_SIZE <- 13
DPI_FIG   <- 300

CLUSTER_LEVELS_FIG <- CLUSTER_LEVELS

# ---- Run figure scripts -----
# Each file should:
# - assume song_df / pca_core / pca_variance / pca_loadings exist
# - build p_XX
# - save to DIR_FIG
# - not read data again

source("R/figures/F1_pca_loadings.R")
source("R/figures/F2_cluster_persistence.R")
source("R/figures/F3_cluster_rankings.R")
source("R/figures/F4_cluster_duration.R")

message("Figures complete. Outputs written to outputs/figures/")

```

```

# R/helpers.R -----
# Shared plot styling, project constants, and small reusable utilities

suppressPackageStartupMessages({
  library(ggplot2)
  library(dplyr)
  library(readr)
  library(scales)
})

options(
  pillar.min_chars = 6,
  pillar.max_width = Inf,
  pillar.max_extra_cols = Inf
)

# ---- Analysis spec -----
# Centralise fixed settings to keep EDA + figure scripts consistent.

SEED <- 123

K_PCS_CLUSTER <- 5 # number of PCs used for k-means
K_CLUSTERS_FINAL <- 5 # chosen number of clusters (k)
K_RANGE_DIAG <- 2:10 # k range for elbow/silhouette diagnostics

N_SIL_SAMPLE <- 4000 # silhouette subsample size (prevents dist() O(n^2) hang)

KMEANS_NSTART <- 50 # k-means initialisations (stability)

# ---- Variables -----

vars_core <- c(
  "acousticness",
  "danceability",
  "energy",
  "valence",
  "tempo",
  "speechiness",
  "liveness",
  "instrumentalness"
)

pop_vars <- c("weeks_on_chart", "song_rank_peak", "song_rank_entry",
  "song_rank_final")

# ---- Cluster labels + colours -----
# Cluster IDs in your pipeline are assumed to be 1..5 (k-means output).
# We apply semantic labels here so every script is consistent.

```

```

CLUSTER_LEVELS <- c(
  "Melodic-positive",
  "High-tempo vocal",
  "Speech-dominant",
  "Instrumental-dominant",
  "Mellow acoustic"
)

CLUSTER_PAL <- setNames(
  hue_pal(l = 65, c = 100)(5),
  CLUSTER_LEVELS
)

apply_cluster_labels <- function(df) {
  df %>%
    mutate(
      cluster = factor(
        cluster,
        levels = seq_along(CLUSTER_LEVELS),
        labels = CLUSTER_LEVELS
      )
    )
}

scale_fill_cluster <- function(...) {
  scale_fill_manual(values = CLUSTER_PAL, breaks = CLUSTER_LEVELS, drop =
FALSE, ...)
}

scale_colour_cluster <- function(...) {
  scale_colour_manual(values = CLUSTER_PAL, breaks = CLUSTER_LEVELS, drop =
FALSE, ...)
}

# ---- PCA helper: variance explained labels -----

pca_ve_labels <- function(ve, k = 5) {
  k <- min(k, length(ve))
  cum <- cumsum(ve)

  tibble(
    PC = paste0("PC", 1:k),
    ve = ve[1:k],
    cum = cum[1:k],
    label = sprintf("PC%d: %.1f%% (cum %.1f%%)", 1:k, 100 * ve[1:k], 100 * cum[1:k])
  )
}

```

```

# ---- PCA helper: fit + extract core outputs -----
# Pure helper (no IO). Useful to keep PCA outputs consistent across scripts.

fit_pca_core <- function(df, vars = vars_core, k_pcs = K_PCS_CLUSTER) {
  X <- df %>% select(all_of(vars)) %>% as.data.frame()

  pca <- prcomp(X, center = TRUE, scale. = TRUE)

  ve <- (pca$sdev^2) / sum(pca$sdev^2)
  cum_ve <- cumsum(ve)

  k_pcs <- min(k_pcs, ncol(pca$x))

  scores <- as.data.frame(pca$x[, 1:k_pcs, drop = FALSE])
  colnames(scores) <- paste0("PC", seq_len(k_pcs))

  list(
    pca = pca,
    ve = ve,
    cum_ve = cum_ve,
    scores = scores
  )
}

# ---- k-means helper -----
# Ensures deterministic clustering runs across scripts.

run_kmeans <- function(X, k, nstart = KMEANS_NSTART, seed = SEED) {
  set.seed(seed)
  kmeans(X, centers = k, nstart = nstart)
}

# ---- Feature label helpers -----

feature_labels_core <- c(
  acousticness = "Acousticness",
  danceability = "Danceability",
  energy = "Energy",
  instrumentalness = "Instrumentalness",
  liveness = "Liveness",
  speechiness = "Speechiness",
  valence = "Valence",
  tempo = "Tempo"
)

label_features_core <- function(x) {
  unname(feature_labels_core[x])
}

# ---- Directory helper -----

```



```

ensure_dir <- function(path) {
  if (!dir.exists(path)) {
    dir.create(path, recursive = TRUE, showWarnings = FALSE)
  }
}

# ---- Output logging helper -----

with_log <- function(file, expr) {
  ensure_dir(dirname(file))
  sink(file, type = "output")
  on.exit(sink(type = "output"), add = TRUE)
  force(expr)
}

# ---- Log formatting helpers -----

log_h1 <- function(text) {
  cat(
    "\n\n",
    strrep("=", 80), "\n",
    text, "\n",
    strrep("=", 80), "\n",
    sep = ""
  )
}

log_h2 <- function(text) {
  cat(
    "\n",
    text, "\n",
    strrep("-", nchar(text)), "\n",
    sep = ""
  )
}

# ---- Log table helper -----

log_table <- function(x, digits = 3) {
  x <- x |>
  dplyr::mutate(
    dplyr::across(where(is.numeric), ~ round(.x, digits))
  )

  print(
    x,
    n = Inf,
    width = Inf
  )
}

```

```

)
}

# ---- EDA table export helper -----

write_eda_table <- function(x, filename, digits = 3) {
  ensure_dir(dirname(filename))
  x |>
    dplyr::mutate(
      dplyr::across(where(is.numeric), ~ round(.x, digits))
    ) |>
    readr::write_csv(filename)
}

# ---- Theme -----

theme_project <- function(base_size = 12) {
  theme_minimal(base_size = base_size) +
    theme(
      plot.title = element_text(face = "bold"),
      legend.title = element_text(face = "bold"),
      panel.grid.minor = element_blank()
    )
}

# ---- Save helper -----

save_fig <- function(plot, filename, width = 9.5, height = 6.5, dpi = 300) {
  dir.create(dirname(filename), showWarnings = FALSE, recursive = TRUE)

  ggsave(
    filename = filename,
    plot = plot + theme_project(),
    width = width,
    height = height,
    dpi = dpi,
    bg = "white"
  )
}

```

```

#
=====
=====
# Figure 01: Acoustic feature profiles by cluster (TRUE RADAR; facets)
#
=====

```

```
=====
```

```
# ---- Local parameters -----
```

```
WIDTH_01 <- 10
```

```
HEIGHT_01 <- 8.6
```

```
Y_MIN_01 <- -1.0
```

```
Y_MAX_01 <- 1.0
```

```
Y_BREAKS_01 <- c(-1.0, -0.5, 0, 0.5, 1.0)
```

```
feature_order <- c(
  "tempo", "danceability", "energy", "valence",
  "acousticness", "instrumentalness", "liveness", "speechiness"
)
```

```
feature_labels <- c(
  tempo      = "Tempo",
  danceability = "Dance",
  energy      = "Energy",
  valence     = "Valence",
  acousticness = "Acoustic",
  instrumentalness = "Instrumentalness",
  liveness    = "Live",
  speechiness  = "Speechiness"
)
```

```
# ---- Data preparation -----
```

```
cluster_profile_df <- read_csv(
  "clean/cluster_feature_profiles.csv",
  show_col_types = FALSE
)
```

```
profile_df_01 <- cluster_profile_df %>%
  mutate(
    feature      = factor(feature, levels = feature_order),
    idx          = as.integer(feature),
    mean_value_plot = pmin(pmax(mean_z, Y_MIN_01), Y_MAX_01)
  ) %>%
  arrange(cluster, idx)
```

```
# ---- True radar geometry -----
```

```
K_01 <- length(feature_order)
```

```
R_MAX <- Y_MAX_01 - Y_MIN_01      # total radial span
```

```
R_RINGS <- (Y_BREAKS_01 - Y_MIN_01) # include z = 0 ring
```

```
profile_xy_01 <- profile_df_01 %>%
```

```

mutate(
  theta = pi/2 - 2*pi*(idx - 1)/K_01,
  r     = mean_value_plot - Y_MIN_01,    # non-negative radius
  x     = r * cos(theta),
  y     = r * sin(theta)
) %>%
arrange(cluster, idx)

# Close each polygon explicitly
path_xy_01 <- profile_xy_01 %>%
  group_by(cluster) %>%
  arrange(idx, .by_group = TRUE) %>%
  dplyr::reframe(
    x = c(x, first(x)),
    y = c(y, first(y))
  )

# Rings (reference circles)
ring_df_01 <- tidyr::expand_grid(
  r = R_RINGS,
  t = seq(0, 2*pi, length.out = 361)
) %>%
mutate(
  x = r * cos(t),
  y = r * sin(t)
)

# Spokes + labels
spokes_df_01 <- tibble(
  feature = factor(feature_order, levels = feature_order),
  idx     = seq_along(feature_order)
) %>%
mutate(
  theta = pi/2 - 2*pi*(idx - 1)/K_01,
  x0     = 0,
  y0     = 0,
  x1     = R_MAX * cos(theta),
  y1     = R_MAX * sin(theta),
  xl     = (R_MAX * 1.14) * cos(theta),
  yl     = (R_MAX * 1.14) * sin(theta),
  lab    = unname(feature_labels[as.character(feature)])
)

# Canvas limits (protect label space)
LIM <- R_MAX * 1.22

# ---- Plot -----

p_01 <- ggplot() +

```

```

geom_path(
  data = ring_df_01,
  aes(x, y, group = r),
  colour = "grey82",
  linewidth = 0.75
) +
geom_segment(
  data = spokes_df_01,
  aes(x = x0, y = y0, xend = x1, yend = y1),
  colour = "grey88",
  linewidth = 0.6
) +
geom_polygon(
  data = path_xy_01,
  aes(x, y, group = cluster, fill = cluster),
  alpha = 0.25
) +
scale_fill_cluster() +
geom_path(
  data = path_xy_01,
  aes(x, y, group = cluster, colour = cluster),
  linewidth = 1.35,
  linejoin = "mitre"
) +
geom_point(
  data = profile_xy_01,
  aes(x, y, colour = cluster),
  size = 2.8
) +
geom_text(
  data = spokes_df_01,
  aes(x = x1, y = y1, label = lab),
  size = 3.2
) +
coord_fixed(
  xlim = c(-LIM, LIM),
  ylim = c(-LIM, LIM)
) +
facet_wrap(~cluster, ncol = 3) +
scale_colour_cluster() +
labs(
  title = "Acoustic feature profiles by cluster",
  subtitle = "Points show average feature levels for each group. Extreme values are capped
for readability."
) +
theme_project(base_size = BASE_SIZE) +
theme(
  legend.position = "none",
  panel.grid = element_blank(),

```

```

axis.title    = element_blank(),
axis.text     = element_blank(),
axis.ticks    = element_blank(),
strip.text    = element_text(face = "bold", size = 12),
panel.spacing = unit(0.3, "lines")
)

# ---- Save -----

ggsave(
  filename = file.path(DIR_FIG, "01_cluster_profiles.png"),
  plot     = p_01,
  width    = WIDTH_01,
  height   = HEIGHT_01,
  dpi      = DPI_FIG,
  bg       = "white"
)

```

```

# R/figures/F2_cluster_persistence.R -----
#
=====
=====
# Figure 02: Cluster prevalence over time (shares by year; line chart)
# Output: 02_cluster_persistence.png
#
=====
=====

# ---- Local parameters -----

WIDTH_02 <- 13.5
HEIGHT_02 <- 7.8

YEAR_MIN_02 <- 1964
YEAR_MAX_02 <- 2018

# ---- Data preparation -----

ts_df_02 <- song_df %>%
  tidyr::separate_rows(years_on_chart, sep = ";") %>%
  mutate(
    year = as.integer(years_on_chart),
    cluster = factor(cluster, levels = names(CLUSTER_PAL))
  ) %>%
  filter(year >= YEAR_MIN_02, year <= YEAR_MAX_02) %>%
  count(year, cluster, name = "n_songs") %>%

```

```

group_by(year) %>%
mutate(share = n_songs / sum(n_songs)) %>%
ungroup()

# ---- Plot -----

p_02 <- ggplot(ts_df_02, aes(x = year, y = share, colour = cluster)) +
  geom_line(linewidth = 1.6, lineend = "round") +
  scale_x_continuous(
    breaks = seq(YEAR_MIN_02, YEAR_MAX_02, by = 10),
    expand = expansion(mult = c(0.01, 0.01))
  ) +
  scale_y_continuous(
    labels = scales::percent_format(accuracy = 1),
    limits = c(0, 0.6),          # cap at 60%
    breaks = seq(0, 0.6, by = 0.1),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_colour_manual(
    values = CLUSTER_PAL,
    drop   = FALSE
  ) +
  labs(
    title   = "Cluster prevalence over time",
    subtitle = "Yearly share of charting songs",
    x       = "Year",
    y       = "Share of songs"
  ) +
  theme_project(base_size = BASE_SIZE) +
  theme(
    panel.grid.minor = element_blank(),
    legend.title     = element_blank(),
    legend.position  = "right"
  )

# ---- Save -----

ggsave(
  filename = file.path(DIR_FIG, "02_cluster_persistence.png"),
  plot     = p_02,
  width    = WIDTH_02,
  height   = HEIGHT_02,
  dpi      = DPI_FIG,
  bg       = "white"
)

```

```
#
```

```

=====
=====
# Figure 03: Median peak chart rank by cluster-year (entry year)
#
=====
=====

# ---- Local parameters -----

WIDTH_03 <- 10
HEIGHT_03 <- 6.8

POINT_ALPHA <- 0.95
POINT_SIZE <- 1.60
SWARM_WIDTH <- 0.35 # horizontal spread within cluster

# ---- Data prep -----

df_03 <- song_df %>%
  filter(!is.na(cluster), !is.na(song_rank_peak), !is.na(entry_year)) %>%
  mutate(
    cluster = factor(cluster, levels = CLUSTER_LEVELS),
    entry_year = as.integer(entry_year),

    # Convert reversed rank back to chart rank (1 = best)
    peak_rank_chart = 101 - song_rank_peak
  ) %>%
  group_by(cluster, entry_year) %>%
  summarise(
    peak_rank_med = median(peak_rank_chart),
    .groups = "drop"
  )

# Zoom to observed yearly medians (+ small buffer)
y_min <- min(df_03$peak_rank_med, na.rm = TRUE) - 3
y_max <- max(df_03$peak_rank_med, na.rm = TRUE) + 3

# Per-cluster summary across years (median + IQR of yearly medians)
sum_03 <- df_03 %>%
  group_by(cluster) %>%
  summarise(
    med = median(peak_rank_med),
    q25 = quantile(peak_rank_med, 0.25),
    q75 = quantile(peak_rank_med, 0.75),
    .groups = "drop"
  )

# ---- Plot -----
# Requires ggbeeswarm

```



```

p_03 <- ggplot(df_03, aes(x = cluster, y = peak_rank_med, colour = cluster)) +
  ggbeeswarm::geom_quasirandom(
    width = SWARM_WIDTH,
    groupOnX = TRUE,
    alpha = POINT_ALPHA,
    size = POINT_SIZE
  ) +
  geom_errorbar(
    data = sum_03,
    aes(x = cluster, ymin = q25, ymax = q75),
    inherit.aes = FALSE,
    width = 0.20,
    linewidth = 0.9,
    colour = "black"
  ) +
  geom_point(
    data = sum_03,
    aes(x = cluster, y = med),
    inherit.aes = FALSE,
    size = 3.0,
    colour = "black"
  ) +
  scale_y_reverse(
    breaks = seq(
      floor(y_min / 10) * 10,
      ceiling(y_max / 10) * 10,
      by = 10
    ),
    expand = c(0, 0)
  ) +
  coord_cartesian(ylim = c(y_max, y_min)) +
  scale_colour_cluster() +
  labs(
    title = "Peak chart rank by acoustic structure",
    subtitle = "Points are yearly medians (entry year). Black dot shows the median across
years; black bar shows the IQR (25th–75th percentile).",
    colour = "Acoustic cluster",
    x = NULL,
    y = "Peak chart rank (1 = best)"
  ) +
  theme_project(base_size = BASE_SIZE)

# ---- Save -----

save_fig(
  p_03,
  file.path(DIR_FIG, "03_cluster_rankings.png"),
  width = WIDTH_03,

```

```
height = HEIGHT_03,  
dpi    = DPI_FIG  
)
```

```
#  
=====
```

```
# Figure 04: Song longevity by cluster (ridgeline; raw scale)  
#  
=====
```

```
suppressPackageStartupMessages({  
  library(dplyr)  
  library(ggplot2)  
  library(ggribes)  
})  
  
source("R/helpers.R")  
  
WIDTH_04 <- 12  
HEIGHT_04 <- 7  
  
# ---- Data preparation -----  
  
df_04 <- song_df %>%  
  { if (is.numeric(. $cluster) || is.integer(. $cluster)) apply_cluster_labels(.) else . } %>%  
  mutate(cluster = factor(cluster, levels = CLUSTER_LEVELS)) %>%  
  filter(!is.na(cluster), !is.na(weeks_on_chart)) %>%  
  filter(is.finite(weeks_on_chart), weeks_on_chart > 0)  
  
x_p99 <- as.numeric(  
  quantile(df_04$weeks_on_chart, 0.99, na.rm = TRUE)  
)  
  
# ---- Plot -----  
  
p_04 <- ggplot(df_04, aes(x = weeks_on_chart, y = cluster, fill = cluster)) +  
  geom_density_ridges(  
    scale = 1,  
    alpha = 0.4,  
    colour = "white",  
    linewidth = 0.25  
  ) +
```

```

coord_cartesian(xlim = c(0, x_p99)) +
scale_fill_cluster() +
theme_project(base_size = BASE_SIZE) +
theme(legend.position = "none") +
labs(
  title = "Acoustic Structure Chart longevity",
  subtitle = "Ridgeline densities of song chart duration (trimmed at 99th percentile)",
  x = "Weeks on chart",
  y = "Acoustic Structure"
)

# ---- Save -----

ensure_dir(DIR_FIG)

ggsave(
  filename = file.path(DIR_FIG, "04_cluster_duration.png"),
  plot     = p_04,
  width    = WIDTH_04,
  height   = HEIGHT_04,
  dpi      = DPI_FIG,
  bg       = "white"
)

```